SETTING UP A COMPANY LAN NETWORK WITH AN APACHE WEB SERVER

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ROUTER CONFIGURATION

- R1 was configured to connect to PC1, PC2, Kali Linux machine, and S1 on a separate subnet
- The subnet for these devices is 192.168.10.0/24
- Interface E0/1 was connected to R2
- R1 was given sub-interface E0/0.10 to connect with S1 with the address of 192.168.10.65
- OSPF routing protocol was configured to connect to R3



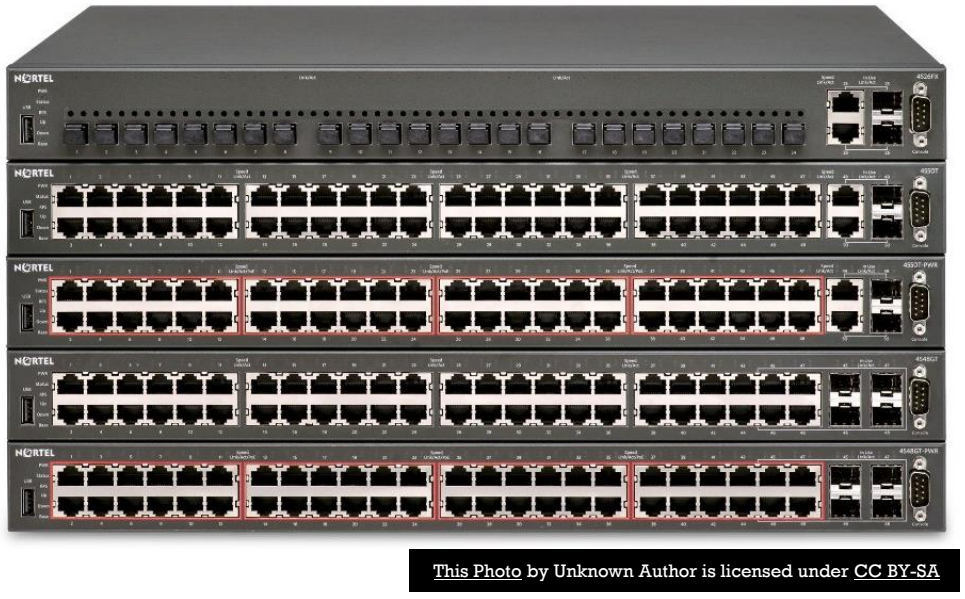
ROUTER CONFIGURATION

- Router 2 (R2) was configured to connect R1 to Router 3 (R3)
- OSPF routing protocol was configured on R2 to connect the routers
- Interface E0/0 was connected to R1 with an address of 192.168.20.1
- Interface E0/1 was connected to R3 with an address of 192.168.30.1
- There is a default static route from R2 to the internet via 192.168.83.2 that is attached to E0/2
- R2 included an extended access-list (ACL) to allow the subnets of R1, R2, and R3 to connect to the internet
  - Access-list 100

ROUTER CONFIGURATION

- R3 is on a different subnet than R1
- The subnet is 192.168.40.0/24
- The Ubuntu server is attached to R3
- Interface E0/0 is connected to R2 with an address of 192.168.30.2

SWITCH CONFIGURATION



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- S1 was configured to allow PC1, PC2, and Kali Linux to access VLAN10
- VLAN10 was given the address 192.168.10.66
- Port-security was enabled on all switch ports with access to VLAN10
- Interfaces E0/0-2 was connected to each of the end devices
- Trunking was configured on interface E0/3 to trunk VLAN10 to R1

PC CONFIGURATION



- PC1 was configured to connect to VLAN10 via S1
- PC1 was given the address of 192.168.10.67/29
- PC2 was configured to connect to VLAN10 via S1
- PC2 was given the address of 192.168.10.68/29

UBUNTU SERVER

- The Ubuntu Server was configured on the same subnet as R3
- Ubuntu Server was given the address 192.168.40.100/24
- The server would run the apache2 service to create a web page for the company network to display information
- The Ubuntu server was also configured to run an SSH and FTP server for the network



UBUNTU SERVER CVES

CVE-2019-0220

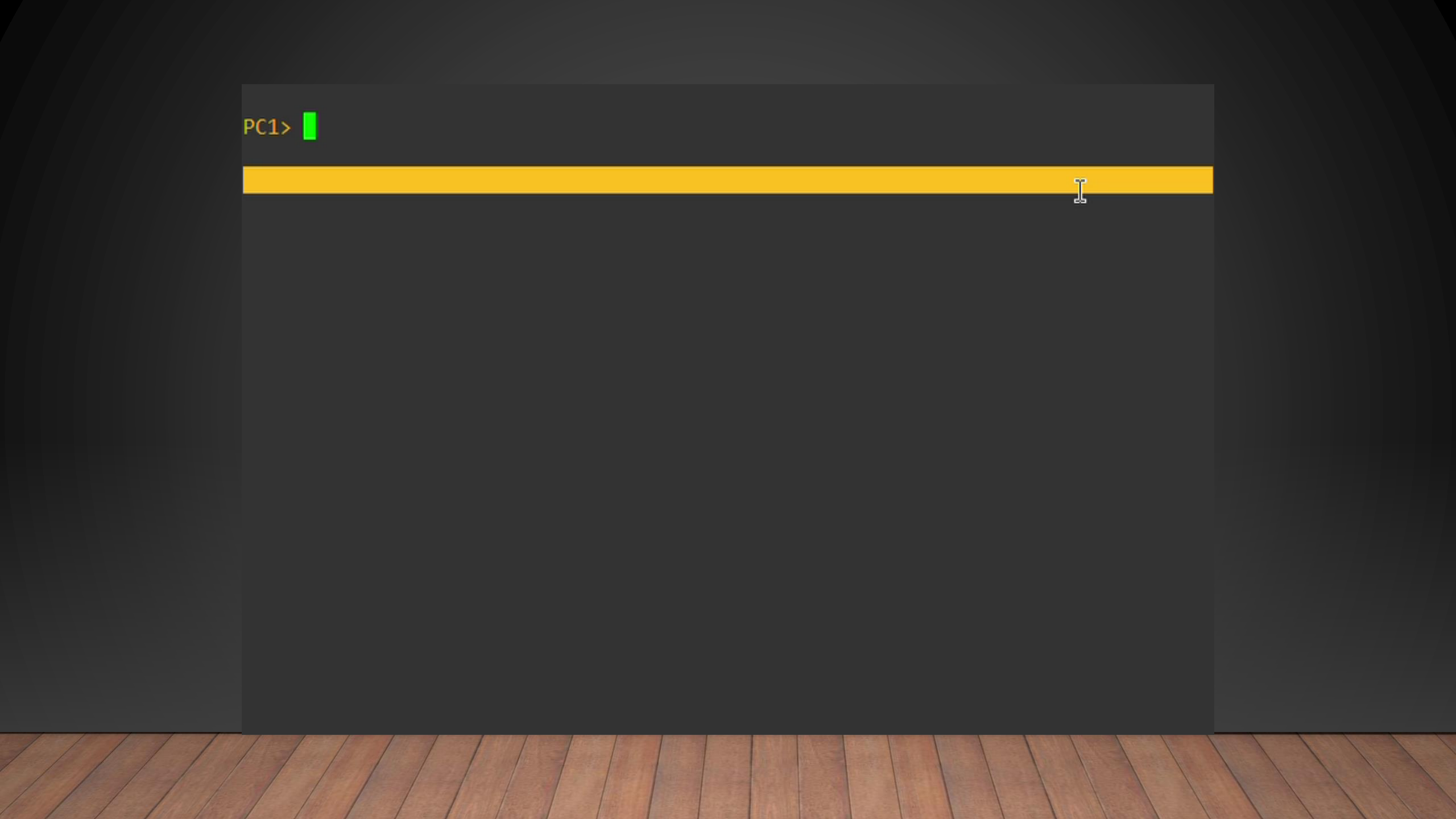
- A vulnerability was found in Apache HTTP Server 2.4.0 to 2.4.38. When the path component of a request URL contains multiple consecutive slashes ('/'), directives such as LocationMatch and RewriteRule must account for duplicates in regular expressions while other aspects of the servers processing will implicitly collapse them.

CVE-2020-1934

- In Apache HTTP Server 2.4.0 to 2.4.41, mod_proxy_ftp may use uninitialized memory when proxying to a malicious FTP server.

CVE-2022-2322

- USBView 2.1 before 2.2 allows some local users (e.g., ones logged in via SSH) to execute arbitrary code as root because certain Polkit settings (e.g., allow_any=yes) for pkexec disable the authentication requirement. Code execution can, for example, use the --gtk-module option. This affects Ubuntu, Debian, and Gentoo.


PC1>

KALI LINUX

- The Kali Linux machine was configured to connect to VLAN10 via S1
- The Kali Linux machine was given the address of 192.168.10.69/29
- John the Ripper, Nmap, and Metasploit were tools used for penetration (pen) testing



NMAP

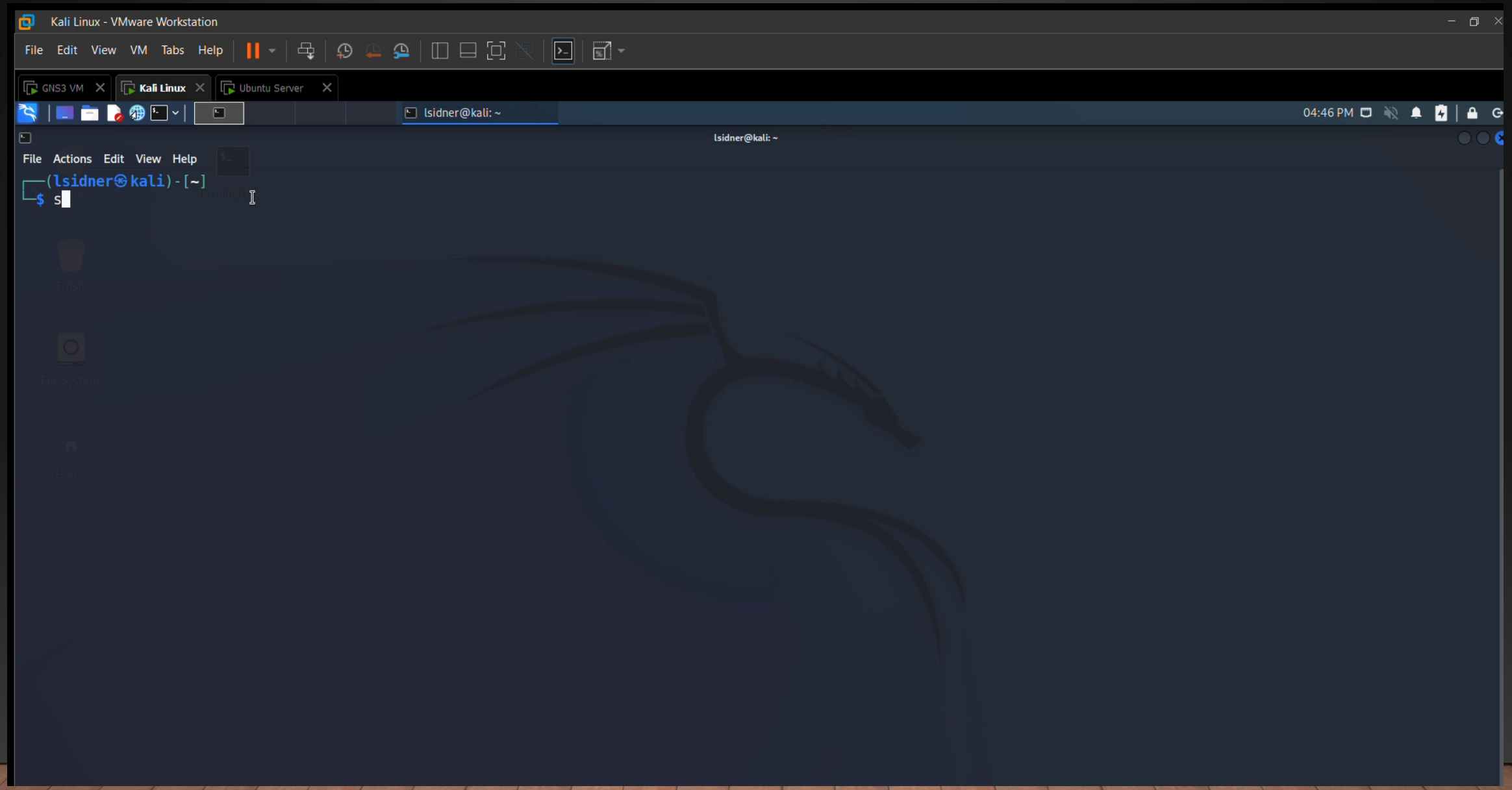
The Nmap tool was used to scan the subnet of R3

Nmap returned ports 21, 22, and 80 were open on host 192.168.40.100; which is the address of the Ubuntu server

Port 21 = File Transfer Protocol (FTP)

Port 22 = Secure Shell (SSH)

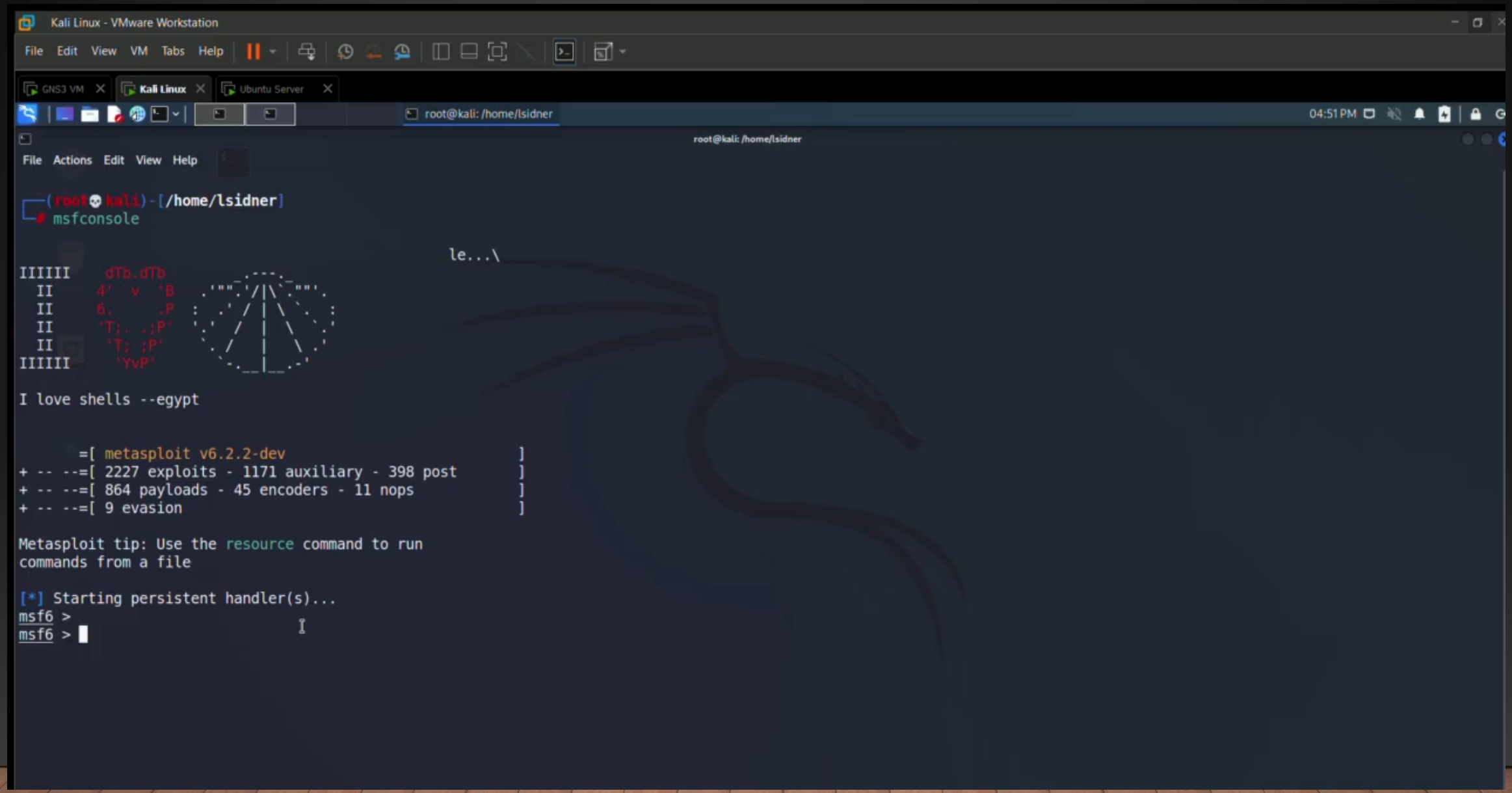
Port 80 = Hypertext Transfer Protocol (HTTP)


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
GNS3 VM
Kali Linux
Ubuntu Server
lsidner@kali: ~
04:46 PM
lsidner@kali: ~
File Actions Edit View Help
(lsidner㉿kali)-[~]
$ s
Trash
File System
Home



METASPLOIT

- The Metasploit tool was used to sniff passwords on the command line
  - The password sniffer was psnuffle which is located in the auxiliary/sniffer/psnuffle folder in MSF console
- The password and username for the Ubuntu server was found by logging in via FTP.
- To retrieve the password file, SSH login was used.
  - The password was entered to log into the machine


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
GNS3 VM
Kali Linux
Ubuntu Server
root@kali: /home/lsidner
04:51 PM
root@kali: /home/lsidner
File Actions Edit View Help
(root kali)-[/home/lsidner]
msfconsole
le...\
IIIIII dTb.dTb
II 4' v 'B
II 6. .P
II 'T; .;P'
II 'T; ;P'
IIIIII 'YvP'
I love shells --egypt
=[metasploit v6.2.2-dev]
+ -- --=[2227 exploits - 1171 auxiliary - 398 post]
+ -- --=[864 payloads - 45 encoders - 11 nops]
+ -- --=[9 evasion]
Metasploit tip: Use the resource command to run commands from a file
[*] Starting persistent handler(s)...
msf6 >
msf6 >

ESCALATING PRIVILEGES

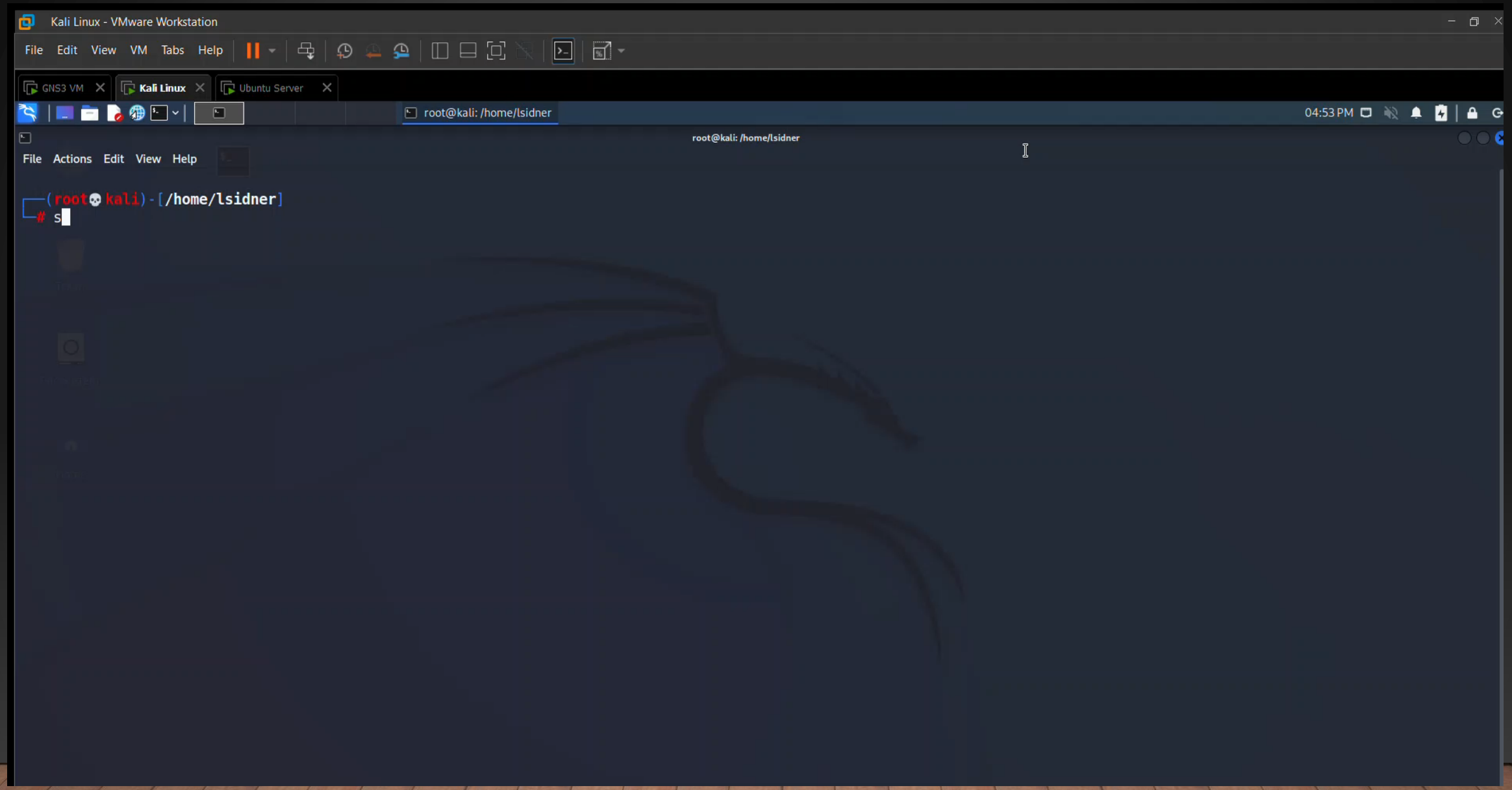
Since the user of the Ubuntu server is a sudo user, escalating privileges was used by typing sudo su root

From there, locating the /etc/shadow file was easy

The cp command was used to copy the file to the home directory of the user (/home/user)

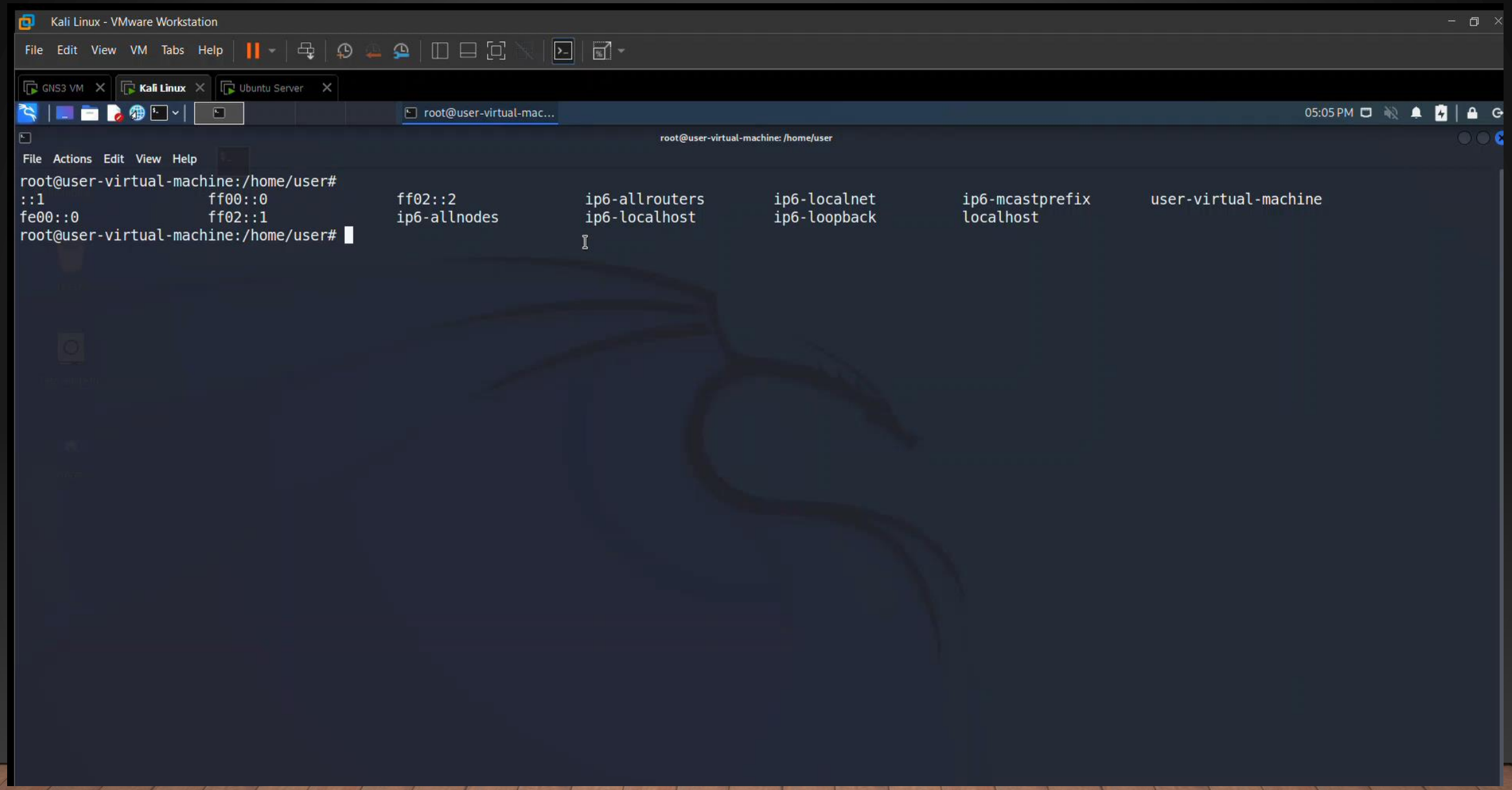
- This command will also be used to change the name of the copied file

The permissions were changed using chmod and chown commands to allow for reading of the file without being the root user


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
GNS3 VM
Kali Linux
Ubuntu Server
root@kali: /home/lsidner
04:53 PM
root@kali: /home/lsidner
File Actions Edit View Help
(root kali)-[/home/lsidner]
s

DOWNLOADING THE FILE

- Now, log in to the ftp server using the login credentials given
- Download the shadow file
- The get command will send the file to the Kali Linux system
  - Get shadow.txt (or whatever given file name)


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
GNS3 VM
Kali Linux
Ubuntu Server
root@user-virtual-mac...
05:05 PM
root@user-virtual-machine: /home/user
File Actions Edit View Help
root@user-virtual-machine:/home/user#
::1 ff00::0 ff02::2 ip6-allrouters ip6-localnet ip6-mcastprefix user-virtual-machine
fe00::0 ff02::1 ip6-allnodes ip6-localhost ip6-loopback localhost
root@user-virtual-machine:/home/user#

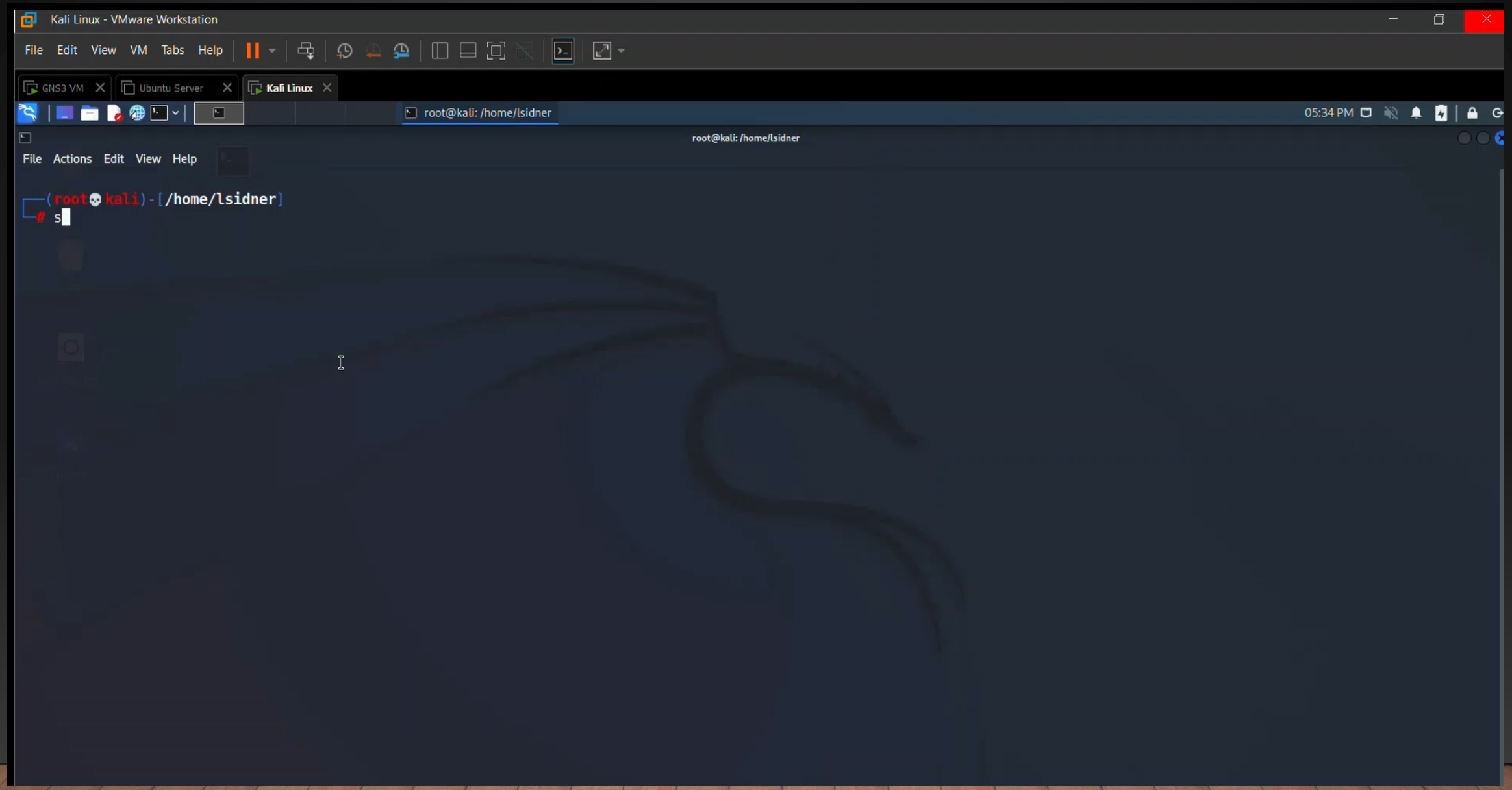
JOHN THE RIPPER

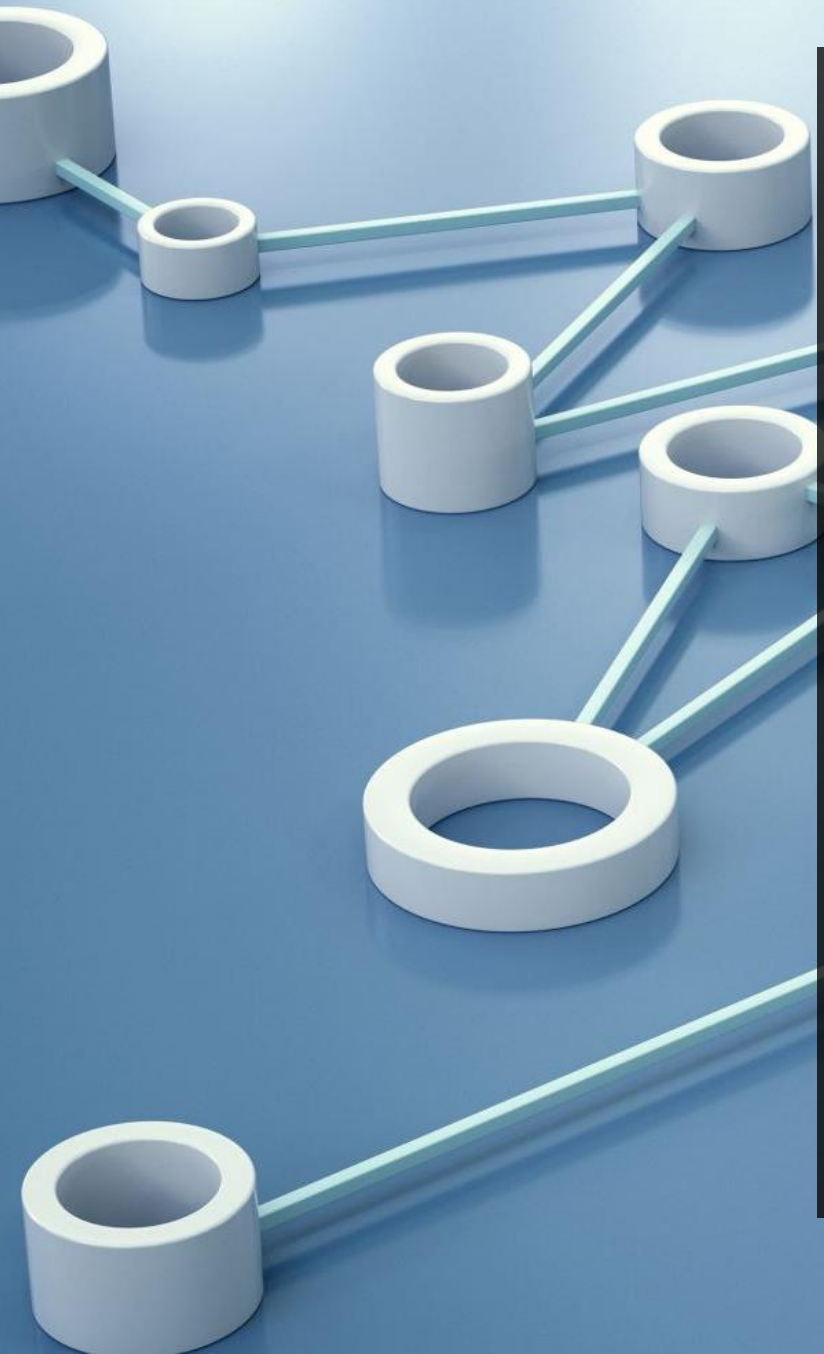
Next, use John the Ripper to crack the password file.
> John shadow.txt

The password for the root user will be shown in plaintext. It will show the username and password.

This process takes a while to use as it exhausts every possible password using a brute-force method




Kali Linux - VMware Workstation
File Edit View VM Tabs Help
GNS3 VM
Ubuntu Server
Kali Linux
root@kali: /home/lsidner
05:34 PM
root@kali: /home/lsidner
File Actions Edit View Help
(root kali)-[/home/lsidner]
s



TROUBLESHOOTING

- One of the issues that may occur is the ability to connect to the internet
  - To solve this, a cloud was added to the topology that was connected to R2 via interface E0/2
  - R2 was given an IP address through DHCP and a default route to the internet
- Another issue was connecting to the GNS3 VM
  - To solve this, go to Edit > Preferences > Server to change the server preferences
  - Enable local server to run the GNS3 VM from the current machine (i.e., computer)
  - Use the loopback address of 127.0.0.1 and port 3080 TCP for best results

REFERENCES

Andrea, H. (2019, November 17). *Configuring NAT on Cisco Routers Step-by-Step (PAT, Static NAT, Port Redirection)*. Networks Training. https://www.networkstraining.com/configuring-nat-on-cisco-routers/

Bagci, T. (2020, January 15). *How to Install Cisco IOU on GNS3 VM*. SYSNETTECH Solutions. https://www.sysnettechsolutions.com/en/install-cisco-iou-gns3-vm/

ComputerNetworkingNotes. (2019, December 21). *How to configure Default Routing in Cisco Routers*. https://www.computernetworkingnotes.com/ccna-study-guide/how-to-configure-default-routing-in-cisco-routers.html

CVE-2019-0220. (2018, November 14). CVE. https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2019-0220

CVE-2020-1934. (2019, December 2). CVE. https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2020-1934

CVE-2022-23220. (2022, January 14). CVE. https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2022-23220

GNS3 2.0.2 VMware connect cloud to Internet. (2017, June 11). YouTube. https://www.youtube.com/watch?v=KSHJrHp2WTo

K. (2021a, July 10). *How to Use John the Ripper | John the Ripper Password Cracker*. TECHOFIDE. https://techofide.com/blogs/how-to-use-john-the-ripper-john-the-ripper-password-cracker-techofide/

L. (2021b, June 16). *vsftpd - Allow root User Login (ftp server - ubuntu)*. YouTube. https://www.youtube.com/watch?v=glcO2zhG94Q&feature=youtu.be

L. (2022, April 15). *How to Install and Run A GNS3 VM [A Step-by-Step Guide]*. MiniTool. https://www.minitool.com/partition-disk/gns3-vm.html

Offensive Security. (2019, October 31). Password Sniffing - Metasploit Unleashed. https://www.offensive-security.com/metasploit-unleashed/password-sniffing/

S. (2020, March 29). *Configure Inter VLAN Routing on Cisco Router and Layer 3 Switches*. TECHNIG. https://www.technig.com/configure-inter-vlan-routing-cisco-router/